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On the Generation of Alternatives in Decision Analysis Problems

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Our primary purpose in this paper is to identify *options generation* as an extremely important part of the decision-making task. We review the behavioural psychology and computer science literature, as well as the decision sciences literature, and discover an apparent lack of research on the problem. However, we are able to discern some important issues and use these to guide us in our attempt to define a framework for options generation. We do not attempt to devise an automatic options generation procedure, but focus on those factors that prevent the decision maker from constructing a rich set of decision alternatives. By doing this we are able to prescribe a prototype option prompting environment.

INTRODUCTION

DECISION ANALYSIS is the systematic application of decision theory to problems of choice between alternatives. As such, it has developed over the last two decades into a mature and powerful approach. It is able to handle a large number of problems, and there are many well developed techniques for assisting the decision maker in choosing among competing alternatives; for examples see Raiffa,[1] Barclay *et al.*[2] and Holloway.[3]

In practice, however, a decision analyst aids the decision maker in specifying the alternatives (options generation) as well as in assessing the value of these alternatives (choice resolution). It is somewhat puzzling, then, to find that while "Practised decision analysts... report that a major part of many studies is the specification of the set of alternative courses of action" (Watson and Brown[4]) a search of the decision sciences literature indicates an almost complete lack of interest in problem specification.

It has been suggested that the reason for this is that decision scientists view options generation as an act of creative insight and thus not addressable by normative techniques. Such an attitude would seem somewhat shortsighted. The psychology literature is replete with studies which show that, even in the most favorable circumstances, human decision makers exhibit suboptimal behaviour and are prone to focus on a narrow range of alternatives. A practically relevant theory should address all aspects of the problem, not just those which are mathematically tractable.

Although the decision sciences literature does not address the options generation problem in any significant way, there are some relevant studies in other disciplines. So, for example, the recent work by Pitz *et al.*[5] is an attempt to understand the influence of information conditions on an individual's ability to generate options, and their results clearly indicate that focusing on the decision maker's objectives is the most successful way of generating choices. There has also been considerable research on the effects of cognitive biases (see Slovic *et al.*[6] and Einhorn and Hogarth[7] for reviews) and individual decision style (MacCrimmon and Taylor[8]) on the decision maker's ability to use and analyse information.

The purpose of this paper is to explore the options generation problem. Since the decision maker cannot make an informed choice unless he is aware of all the relevant alternatives, we are interested in ways of ensuring that he does consider a rich set of decision options. Our investigation draws upon work in the psychology of decision making and previous attempts to construct computer-based decision aids. The goal is not to provide a definitive solution to the problem, but to examine those factors which most

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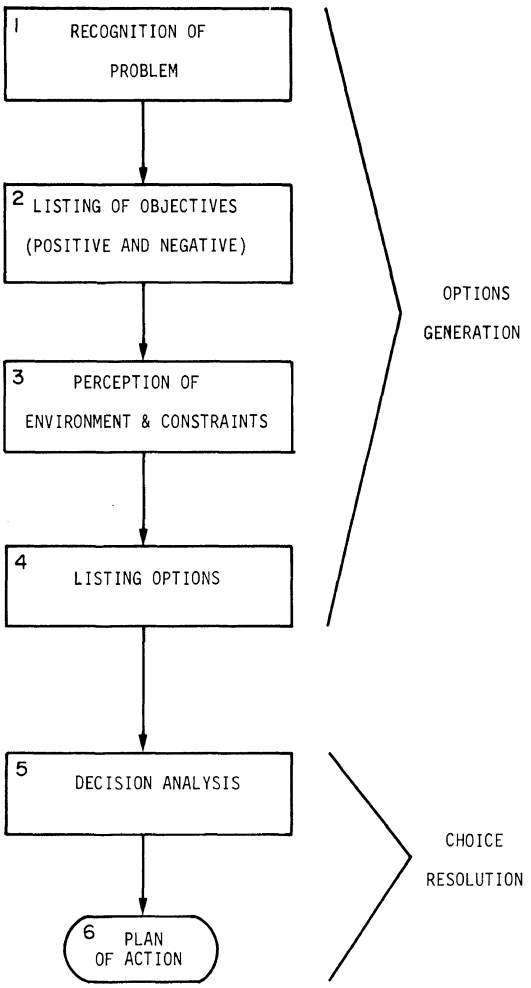


FIG. 1. *The decision making process.*

affect options generation and then to suggest a paradigm for aiding the decision maker in this difficult task.

In the next section of the paper we define the problem more clearly. We draw together our impressions of the literature and use them to guide us in defining a unified structure for options generation. Our solution is a two phase *iterative* structure that emphasizes *goals* rather than sequences of actions. It is inherently inter-active, and its characteristic feedback mechanisms help ensure a stable, rich set of alternatives. We conclude with a simple example that illustrates some of the important features of our approach.

A UNIFIED STRUCTURE FOR OPTIONS GENERATION

The decision making process consists of a sequence of steps that starts with a recognition of the problem and concludes with a plan of action. A schematic view of this is shown in Fig. 1. Steps 5 and 6 are what we have called choice resolution and are the domain of the majority of work in the decision sciences. Steps 1–4 are the options generation phase, and in this paper we are concerned with an investigation of those factors which influence the decision maker's ability to generate appropriate sets of alternatives.

Our literature review has uncovered several important factors that influence options generation. The psychological studies all attest to their impact, and the computer science studies all address at least some of them. However, no one effort has succeeded in drawing together all these factors into a unified approach.

A systematic effort to tackle the options generation problem must recognize the following issues:

—the importance of goals and objectives;
—the value of structuring the major decision influences;
—the impact of questioning and framing;
—the ability to highlight critical events;
—the essential nature of iteration and feedback.

We believe that the methodology we describe in this section is the first integrated attempt to aid the decision maker in generating options. It allows the decision maker to consider each of his objectives in turn, and carefully structures the elements of the decision that influence them. It uses a form of focused devil's advocate to force the decision maker to consider alternative aspects of the problem, and uses sensitivity analysis to concentrate on important events. Finally, it relies on both local and global feedback to ensure a stable and rich view of the options available.

A graphical outline of our approach is shown in Fig. 2. It consists of an evaluation of decision factors (steps 1–4), followed by an assessment of options (steps 5–7). In the first stage we are primarily concerned with elucidating all the relevant decision factors. We organize them hierarchically and then perform an evaluation that indicates the dominant factors with respect to our goals. In the second stage we use the information generated by

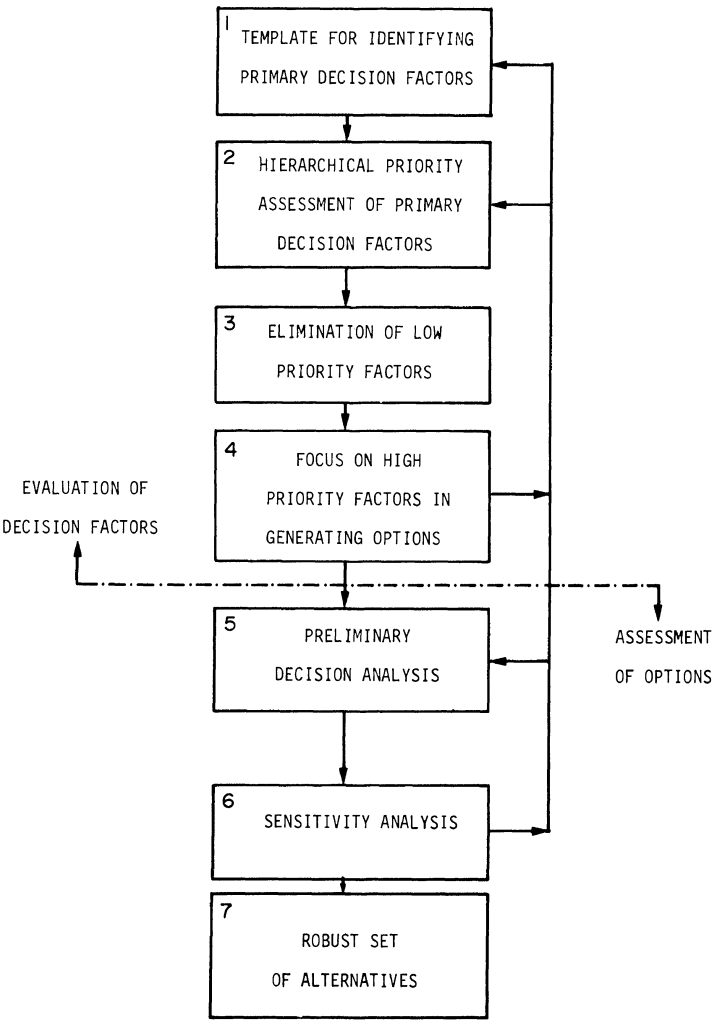


FIG. 2. *Option generation.*

the first stage to develop a preliminary decision analysis. Our purpose here is to examine the implications of the preliminary options, and we are concerned with the sensitivity of this option set to changes in information.

An extremely important feature of this structure is the feedback mechanism, both local and global. We use the sensitivity analysis to highlight critical parts of the problem, and then use this knowledge either to change our interpretation of the high priority factors (local feedback) or to change the structure of the factors (global feedback). The result of this iterative approach is a robust set of alternatives that can be analyzed in detail using the full range of decision analytic tools. In devising the skeleton of an option generation approach, we have introduced five core concepts, namely templates, hierarchical priority assessment, preliminary option generation, sensitivity analysis and feedback. In the remainder of this section we consider each of these in turn.

Templates

The first step in the options generation process involves listing of all decision factors that may have a bearing on the problem and its solution. This templating effort permits structured classification of factors relevant to the problem and its understanding, and we note that this has received some attention from other decision scientists: Robinson,[9] Kelly *et al.*[10]

Our concept of priority assessment requires the construction of a hierarchy of all relevant decision factors. The hierarchy's apex represents the decision maker's overall goal, and the levels below contain elements that the decision maker uses to describe his problem. The order of these levels reflects a logical cause-and-effect relation, and the number of elements in each level is affected by the amount of detail desired (e.g. how many objectives are to be considered).

The hierarchical structure provides a foundation for a generic template ideally suited for our purposes. The general format of one such template is shown in Figure 3. This hierarchical template is generic since its levels are situation-independent. The content of each level will, of course, depend on the particular problem context.

The goal of a decision problem is usually well understood from the context. This may include statements such as: "increase market share", "maintain trade balance", or, in a military context, "resolve the hostage issue". The exogenous decision factors may include such elements as the price of raw materials, the rate of inflation, political activity and other factors that may affect the attainment of the goal.

In considering his competitor's objectives, the decision maker should consider both their positive (i.e. what they are trying to accomplish) and, equally important, their negative (what they are trying to avoid) goals. Competitor's potential actions will include all possible actions that may be required to support the competitor's objectives (both positive and negative).

In considering his own objectives, the decision maker should follow a format similar to that followed for the competitors. In considering his own potential actions, he should explore general arenas in which resources may be mobilized to meet his objectives and counter the competitor's actions. In listing resources he should consider such elements as can support the general actions mentioned above. For example, political IOU's, economic favours, capital equipment, credit facilities, etc.

We should comment here that in actual use, the aid may suggest the general categories to the decision maker and then elicit his input of relevant factors at each level. It is also possible that different decision makers (or experts) may provide input at different levels. For example, the senior management may provide inputs on own objectives, while middle management may provide information on local market conditions.

Hierarchical priority assessment

After templating and the construction of the hierarchy of decision factors, the relative importance of each element in the hierarchy has to be assessed. We feel that the "analytic hierarchy process" introduced by Saaty[11] is the proper tool for this.

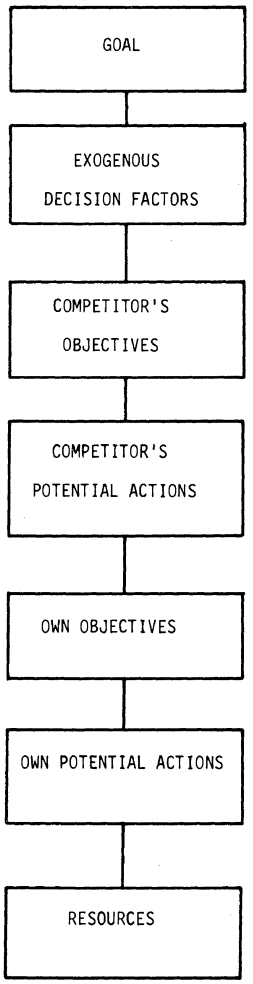


FIG. 3. *Generic hierachy (template).*

As an illustration of how this process works, consider the primitive hierarchy shown in Figure 4. Here the problem is to determine the relative importance of the three activities with respect to the overall goal. We would like the result of this assessment to be a set of weights, w_1, w_2 and w_3, such that $w_1 + w_2 + w_3 = 1$. Saaty's procedure involves a complete pairwise comparison of the activities. The decision maker is asked questions of the following kind:

"Considering activity 1 and activity 2, which one contributes (or dominates) more toward attainment of our goal? What is the strength of contribution (dominance)?"

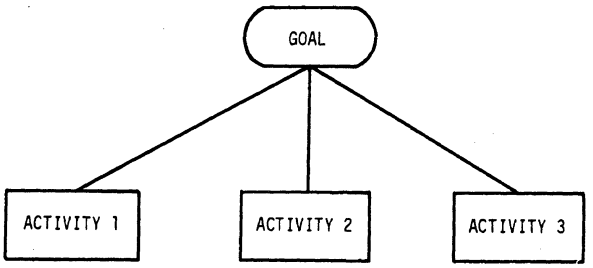


FIG. 4. *An illustrative hierachy.*

The first part of the question involves just an ordinal ranking between the subjects of comparison, while the second part involves a cardinal ranking. To facilitate entering judgments on the "strength of contributing," some scale is needed; such a scale is shown in Table 1.

Questions, such as the one posed above, are asked for all possible pair combinations. Answers are summarized in an $n \times n$ matrix, where n is the number of elements compared ($n = 3$ in the illustrative hierarchy of Figure 4). Each element, a_{ij}, of this matrix shows the dominance of the ith element when compared to the jth element. Thus if activity 1 is judged by the decision maker to 'strongly' dominate activity 2, then a_{12} is set equal to 5. In this model of dominance, we also set a_{21} equal to 1/5. The comparison matrix is thus a reciprocal matrix, and as a result we can recover the weights we require by some simple eigenvalue and eigenvector calculations.

This process is repeated for all levels in the hierarchy and for all elements in these levels, giving a complete set of both 'local' and 'global' priorities. The local priority indicates the relative importance of a particular element to an element in the level immediately above it. The global priorities are computed after the local priorities are given and indicate the impact of the particular element on our overall goal (the apex of the hierarchy).

Preliminary options generation

The result of the previous two steps is a hierarchical description of the factors that influence the decision problem and a set of weights that define the relative priorities of these factors. The options generation aid thus highlights the most important factors and presents them to the decision maker. Because he has been led through a detailed structuring exercise, he should now be in a position to generate a preliminary set of actions that address his basic concerns. It may be the case, however, that although he has developed a more detailed understanding of the problem, he still has very little idea about possible options. If this is so, then some part of the structuring process has to be repeated. In an advanced form of aid it may be possible to do this semi-automatically by using a database of 'likely' options, but in this exploratory discussion we have to assume

TABLE 1. COMPARISON SCALE

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Weak importance of one over another	Experience and judgement slightly favour one activity over another
5	Essential or strong importance	Experience and judgement strongly favour one activity over another
7	Very strong or demonstrated importance	An activity is favoured very strongly over another: its dominance demonstrated in practice
9	Absolute importance	The evidence favouring one activity over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between adjacent scale values	When compromise is needed
Reciprocals of above nonzero	If activity i has one of the above nonzero numbers assigned to it when compared with activity j, then j has the reciprocal value when compared with i	A reasonable assumption

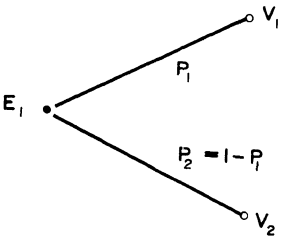


FIG. 5. *A basic probabilistic event.*

that iteration around the templating and prioritization procedures will eventually lead the decision maker to an initial option set.

Once he is in this position, the decision maker can begin to develop, with the help of the aid, a preliminary decision tree. This will allow him to evaluate his set of options and devise a tentative plan of action. That is, once options for action have been generated, their consequences have to be explored. Sensitivity analysis is a way of doing this since it enables the decision maker to identify possible areas of weakness in the plan he is constructing.

Sensitivity analysis

The purpose of sensitivity analysis is to highlight events in the preliminary decision tree that may indicate to the decision maker areas of his problem that are not well understood. The main motivation for this is the belief that the process of discovering sensitive events can help generate new options that will either reduce or circumvent this sensitivity.

To illustrate the point, consider the event shown in Figure 5. Here P_1 is the probability of occurrence of the outcome whose value is V_1 and P_2 is the probability of the outcome whose value is V_2. The expected value is given by

$$E_1 = P_1V_1 + P_2V_2.$$

Since

$$P_2 = 1 - P_1,$$

we have

$$E_1 = P_1(V_1 - V_2) + V_2.$$

This expected value is a linear function of the probability P_1, as shown in Figure 6, for $V_1 > V_2 > 0$. Note that the slope of the expected value line is simply the difference in 'tip values' (i.e. $V_1 - V_2$). Therefore, the bigger the difference the steeper the slope.

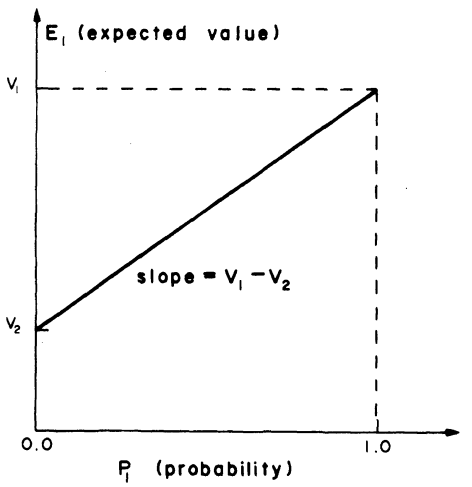


FIG. 6. *Expected value vs probability.*

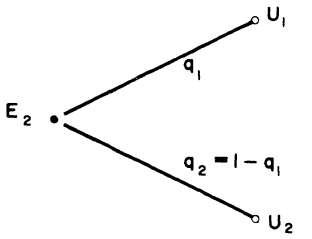


FIG. 7. *A second probabilistic event.*

Consider now another event shown by Figure 7. Here the expected value is similarly given by

$$E_2 = q_1(U_1 - U_2) + U_2.$$

Figure 8 shows the dependence of expected value on probability for both events. We assume that $U_2 < V_1$; the relation of U_2 to V_2 is unimportant for the point of the discussion. It is evident that there are values for P and q that will result in both events having the same expected value and, therefore, being equally preferred. However, at q_1^* event E_2 may be equally preferred to E_1 (assessed with probability of occurrence P_1^*) but any slight assessment error in q_1^* will result in a significant change in expected value for this event due to the steepness of the slope of the E_2 line relative to the E_1 line; i.e. event E_2 is much more sensitive to small assessment errors than is event E_1.

Our approach is to make the decision maker consider those events whose sensitivity to assessment error may significantly affect his plan of action. This is a form of a 'devil's advocate', but instead of looking at every possible event and asking 'what if' questions, we only concentrate on those events that display a sensitive behaviour. Identification of these sensitive events should prompt the decision maker to seek ways of reducing sensitivity. This requires iteration on the basic options set and results either in an enlargement of the set, or modification of it to eliminate sensitive events.

Feedback

The final element in our paradigm is feedback. This forms the important connections between, and within, the two phases of our options generation process. Decision making is rarely, if ever, a one-pass procedure, and so we provide an environment in which the decision maker can review and modify his thinking about the problem.

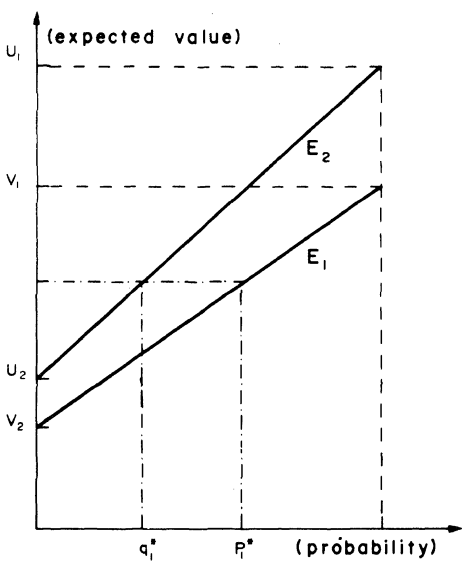


FIG. 8. *Comparison of the two events.*

As we have suggested above, there are three feedback loops within our structure. The first of these provides a mechanism for the decision maker to use the prioritized hierarchy to help him clarify and possibly re-structure the elements of his decision problem (a loop between step 4 and steps 1 and 2 in Figure 2). The second is a way of modifying existing options as a result of sensitivity analysis (a loop between step 6 and step 5). The third involves a significant amount of re-structuring and re-evaluation of the basic decision elements (a loop from step 6 to steps 1 and 2), and we would expect this to happen only in unusual circumstances.

The paradigm thus emphasises the iterative nature of the options generation process. By allowing constant updating and improvement of the decision maker's perception of the problem, we ensure a stable and robust set of decision alternatives.

AN EXAMPLE

In this section we illustrate the first stage of our options generation procedure by applying it to a problem of corporate strategic planning. That is, we show how we can evaluate the decision factors in the planning problem, and how these evaluations can help us generate a preliminary set of planning options.

In our example we imagine a small, high-technology research company that draws on expertise in the decision sciences, computer sciences and systems engineering. At present, the company does basic and applied research for various government agencies, but is concerned with expanding its activities over the medium term future. In particular, the company would like to use some of the results of its various research activities to develop a range of highly competitive products. The purpose of the options generation aid is to help the company think about those factors which influence the future, and to focus as an initial set of strategic plans for achieving its goals.

Figure 9 shows a hierarchy of decision factors that might represent the company's understanding of the problem. The overall goal is to devise a corporate strategic plan for the next three to five years. The first level of the hierarchy is the desired future activities of the company. Here we include product development, software technology, research and consulting. Only product development is an essentially new activity. In the second level, we define some typical problem areas and possible opportunities. We identify manpower, capital resources, market entry, investment risk, government policy and competition as being the important elements. Level three is a list of principal actors: technical staff, management, financial institutions, government and competitors. Finally, in level four we define the possible objectives of the actors: to develop a prestigious reputation, to generate adequate revenue, to have effective control over the operating environment, to capture a respectable market share, and to ensure a high staff morale.

Notice that the connection between levels is sometimes not complete. This shows that the interaction between various elements is considered to be negligible. Thus in pursuing the company's research objectives, only manpower problems, government policy and competition are considered relevant. However, when considering product development, all the elements in level two have some influence.

Obviously, the hierarchy in Figure 9 is an idealized construct. It does, however, convey the idea of using a generic hierarchical structure to express the major goals and requirements of the decision maker (or group of decision makers, as in this case). At this stage we have completed step 1 of the options generation procedure (see Figure 2).

The next task is to assess the relative priorities of these factors (i.e. steps 2 and 3 in Figure 2). The complete sequence of questions and answers is too long to include here, so we restrict ourselves to showing the overall prioritized hierarchy, Figure 10. We notice that in each level there is a clearly dominant factor: product development in level one, market entry in level two, management in level three and reputation in level four. Our initial search for strategic options must focus on these decision factors. However, since we have presented a rather over-simplified view of the problem, it is not realistic to draw anything other than a vague outline of a strategy. Since product development is the

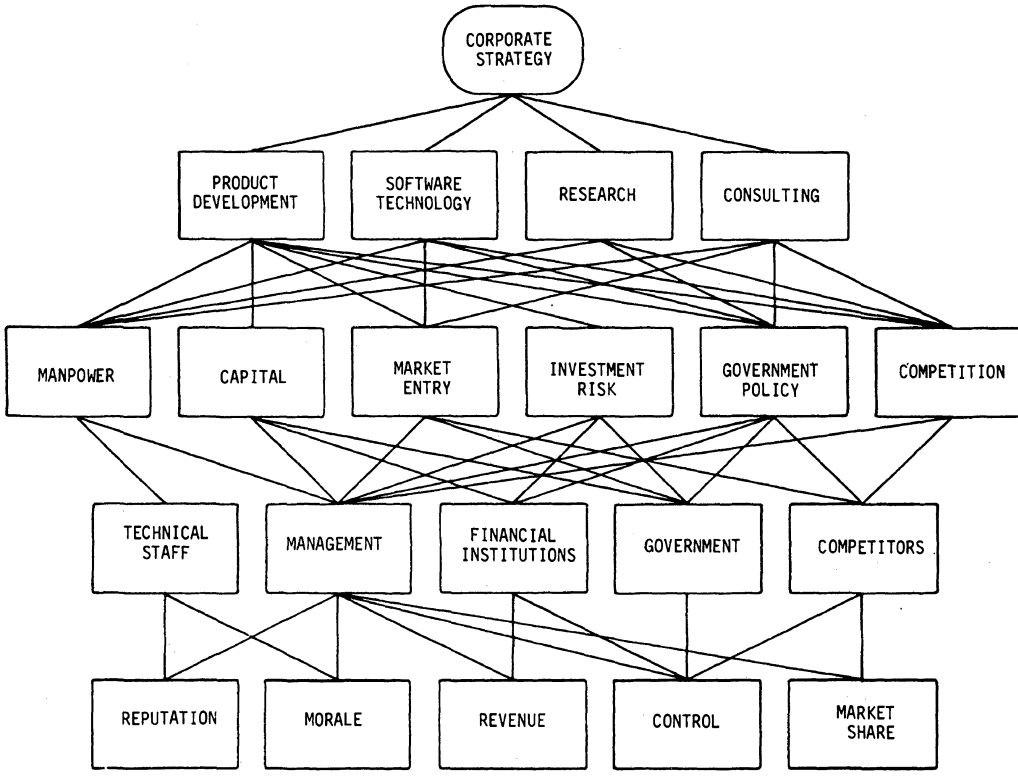


FIG. 9. *Example hierachy.*

TREE STRUCTURE OF : CORPORATE STRATEGIC PLANNING

LEVEL 1 : DESIRED FUTURES

1.1	PRODUCT DEVELOPMENT	0.44	*********
1.2	SOFTWARE TECHNOLOGY	0.11	**
1.3	RESEARCH	0.22	****
1.4	CONSULTING	0.22	****

LEVEL 2 : PROBLEMS/OPPORTUNITIES

2.1	MANPOWER	0.18	****
2.2	CAPITAL	0.14	***
2.3	MARKET ENTRY	0.39	********
2.4	INVESTMENT RISK	0.09	**
2.5	GOVERNMENT POLICY	0.11	**
2.6	COMPETITION	0.09	**

LEVEL 3 : ACTORS

3.1	TECHNICAL STAFF	0.08	**
3.2	MANAGEMENT	0.55	**********
3.3	FINANCIAL INSTITUTIONS	0.15	***
3.4	GOVERNMENT	0.10	**
3.5	COMPETITORS	0.11	**

LEVEL 4 : OBJECTIVES

4.1	REPUTATION	0.37	*******
4.2	REVENUE	0.13	***
4.3	CONTROL	0.21	****
4.4	MARKET SHARE	0.13	***
4.5	MORALE	0.16	***

FIG. 10. *Prioritised hierachy.*

dominant desired activity and since market entry is seen as a significant problem, the company should clearly be thinking in terms of developing a sales force. Management is by far the most important actor in achieving the company's goals, and it is important, therefore, that management skills and attitudes reflect the goals. As a final comment, notice that company reputation is seen to be the most desirable objective. We might interpret this as an indicator of the need to publicize the company in an effective way and to be concerned with the quality of the company's work.

These broad ideas need to be considered in more detail before they could be submitted to any kind of sensitivity analysis. In practice, of course, the hierarchy would be more detailed and more carefully structured and would allow the company to draw up an initial set of planning options.

SUMMARY

Our primary purpose in this paper has been to identify *options generation* as an extremely important part of the decision-making task. We have reviewed the behavioural psychology and computer science literature, as well as the decision sciences literature, and have discovered an apparent lack of research on the problem. However, we have been able to discern some important issues and have used these to guide us in our attempt to define a framework for options generation. We have not attempted to devise an automatic options generation procedure, but have focused on those factors that prevent the decision maker from constructing a rich set of decision alternatives. By doing this we have been able to prescribe a prototype option prompting environment.

Verification of our ideas will come through a series of carefully planned psychological experiments. We are currently devising a suite of computer programs that will allow us to address global questions such as "Does the aid improve options generation?", as well as more detailed questions about the value of various man–machine interface features. Our intention is to start with the problems of templating and priority assessment, and to examine them in terms of their ability to capture important features of the decision maker's knowledge. We shall report on these experiments in a future paper.

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